HOW CANCER CARE IS CHANGING

Treatment protocol is moving from chemo-therapy and radiation to gene- and cell-based therapy, which not only has far less side-effects but also the potential to completely cure cancer

PRECISION ONCOLOGY

This kind of cancer care involves drawing up a treatment plan customised for each patient depending on the type of tumour, the stage and the health condition of the individual. It has widened cancer therapy to include newer generation of targeted therapy options like small molecules and monoclonal antibodies

IMMUNO-ONCOLOGY

Also known as immunotherapy, it involves finding ways to boost the body's own immune system to prevent, control and ultimately eliminate cancer cells. CAR-T therapy and NK Cell therapy are examples of immuno-oncological treatments

MOLECULAR ONCOLOGY

Molecular oncology aims to identify the genes involved in the development of cancer. Treatment for each patient is tailored based on the identification of specific features in tumor biomolecules such as DNA (genomics), RNA (transcriptomics) and protein expression. Unlike invasive biopsies, molecular diagnostics offer a more in-depth insight into cancer and can suggest significantly more precise therapeutic options for patients

WHAT IS CAR-T THERAPY?

Chimeric antigen receptor T cells are T cells that have been genetically modified to produce an artificial T cell receptor for use in immunotherapy. CAR-T therapy uses these cells to reprogramme human T cells to attack cancerous cells in the body

HOW IT IS ADMINISTERED

COLLECTION

A patient's blood is drawn and T cells isolated and frozen to be sent to a CAR-T lab

REPROGRAMMING

CAR-coding viral DNA is incorporated into the human T cells at the lab, transforming them into CAR-T cells

INFUSION

The modified T cells are reintroduced into the patient. They then recognise and destroy cancer cells



CANCERS IT CAN TREAT

Haemotological cancers for now, but products for solid tumours are also under development

COST

Rs 3-4 crore in the US. Expected to cost Rs 30 lakh once the products are made in India

PROGRESS SO FAR

Immuneel has started phase 2/ 3 trials in India, and ImmunoACT has developed its own CAR-T product and started trials for the same. If successful, both these companies will be able to manufacture CAR-T cells in India, bringing down costs significantly and improving accessibility. CAR-T has been shown to completely cure patients of cancer in previous studies

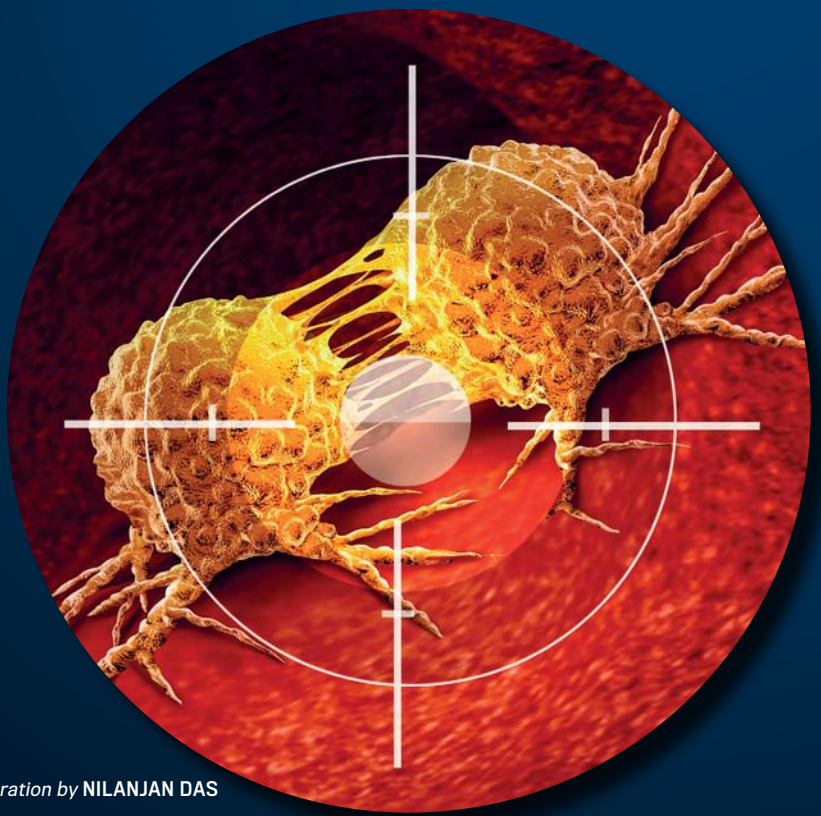


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